MILITARY SPECIFICATION

BED, BUNK, STEEL, SQUARE TUBE, WITH END PANELS, SINGLE AND DOUBLE-DECK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for component materials and their assembly into a single bunk bed with adapters for conversion to double-deck bunk beds.

1.2 Classification. Bunk beds shall be of one type and shall be of the sizes shown on figure 1 and as specified below (see 6.2).

Size 1 - 36-inches wide by 79-inches long overall.

Size 2 - 36-inches wide by 84-inches long overall.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-529 - Enamel, Alkyd, Semi-Gloss.
MMM-A-180 - Adhesive, Polyvinyl Acetate Emulsion.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7105

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MMM-A-188 - Adhesive, Urea Resin Type (Liquid and Powder).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

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MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-151, Supp. A and B - Metals, Test Methods.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, NY 10017).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A208.1 - Mat-formed Wood Particleboard.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A36 - Structural Steel.
ASTM A366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality.
ASTM A513 - Electric-Resistance Welded Carbon and Alloy Steel Mechanical Tubing.
ASTM A569 - Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN WELDING SOCIETY, INC. (AWS)

AWS A3.0 - Terms and Definitions

(Application for copies should be addressed to the American Welding Society, Inc., 2501 Northwest 7th Street, Miami, FL 33125.)

Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The bunk bed shall be of three-piece construction consisting of matching square steel tube, head and foot ends with wood panels, wedge type corner locks each with four bearing points, and a fully-framed spring unit with sinuous (no-sag type) spring fabric, as shown on figures 1, 2, and 3. Bunk beds shall be of two sizes as specified in 1.2.

3.2 Identical items. All units of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations shall be acceptable without prior written approval of the contracting officer.

3.3 First article. Unless otherwise specified (see 6.2), the contractor shall furnish one bunk bed, two beds if adapters are included in the order, for first article inspection and approval (see 4.2.1 and 6.5).

3.4 Materials. Use materials shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Material not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be fabricated from recovered raw materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials reprocessed to become a source of raw materials, as opposed to virgin raw materials which include previously unused metal or metal ore, any undeveloped resource, or new technology which will become source of raw material. Interpret none of the above to mean that the use of used or rebuilt products are allowed, unless otherwise specified.

3.4.1 Steel. All steel shall be smooth, free from rust, scale, pits, scratches, laps and buckles affecting strength and appearance. Unless otherwise specified, the gage numbers or thicknesses specified for steel are subject to standard tolerances. Steel shall be tested in accordance with 4.5.1 where applicable.

3.4.1.1 Steel tubing. Steel tubing for bed ends, adapters, and stretcher braces shall be in accordance with ASTM A513, electric resistance welded type.

3.4.1.2 Steel angles. Steel angles for spring unit frame shall be in accordance with ASTM A36 and shall be 2 by 1-1/2 by 3/16-inches thick. Edges of angles shall be slightly rounded to avoid sharpness.

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3.4.1.3 Spring fabric. Spring fabric shall be high carbon sinuous (no-sag type) furniture spring wire in accordance with Table 7-55 AISI Steel Products Manual. Spring wire shall be No. 10 US Steel Wire Gage, 31-inches long in the flat position, containing 19 full loops including the "2" hook ends. Spring fabric shall be fabricated to 7-1/2-inch diameter plus or minus 1/2 inch. Cut ends shall not have burrs or sharp projections.

3.4.1.4 Spring fabric links. Spring fabric links shall be No. 11 US Steel Wire Gage basic steel wire in accordance with Table 7-48 AISI Steel Products Manual with galvanized finish. Dimensional and size requirements shall be in accordance with figure 4.

3.4.1.5 Helical springs. Helical springs shall be No. 12 US Steel Wire Gage steel premier wire with electrodeposited zinc coating. Tensile strength range in psi shall be 204,000 to 230,000. Dimensional requirements shall be in accordance with figure 4.

3.4.1.6 Corner locks.

3.4.1.6.1 Male corner locks. The male corner locks shall be in accordance with ASTM A569, hot-rolled sheet or strip, commercial quality steel, pickled and oiled of shape, thickness, and dimensional requirements shown on figure 5. As an alternate, the male corner locks shall be in accordance with ASTM A366, cold-rolled sheet steel, commercial quality, and oiled.

3.4.1.6.2 Male corner lock studs. Male corner lock studs shall be AISI Designation C-1010 steel. When upset, shank shall be annealed for riveting with a maximum Rockwell hardness of B-40. When machined, studs shall be AISI Designation CF-B-1113, C-1213, or C-1215 steel. Dimensional requirements shall be as shown figure 5.

3.4.1.6.3 Female corner locks. The female corner locks shall be in accordance with the requirements specified in 3.4.1.6.2 except that the shape, thickness, and dimensional requirements shall be as shown on figure 6.

3.4.2 End panel core. End panel core shall be particleboard in accordance with ANSI 208.1, Grade 1-M-3, weight of 45 pound per cubic foot, and 3/4-inch thick.

3.4.3 High pressure decorative laminated plastic sheet. The high pressure decorative laminated plastic sheet for surfacing end panels shall be in accordance with GP 32 of NEMA LD-3. Pattern, color, and finish shall be as specified (see 3.6.3 and 6.2).

3.4.4 Adhesive. Adhesive for bonding high pressure decorative laminated plastic sheet to panel core shall be in accordance with MMM-A-188 or MMM-A-180.

3.4.5 Paint. The epoxy-powder coating shall meet the requirements of TT-E-529, Type II. Color shall be in accordance with FED-STD-595 Nos. 16251, (Gray gloss), 20059, 20227, 23578, or 27038 as specified (see 3.7.2. and 6.2).

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3.5 Accessories.

3.5.1 End panels. Each bed end panel, two for each bed, shall consist of essentially of a particleboard core (see 3.4.2) surfaced on each side with a high pressure decorative laminated plastic sheet (see 3.4.3). Panel dimensions and thickness shall be as shown on figure 8. Laminated plastic sheets with woodgrain patterns shall have the woodgrain in the horizontal direction, not vertical.

3.5.2 Steel caps. Steel caps for closure of the top of bed posts shall be as shown on figure 7. Caps, four to be provided for each bed, shall be retained within the post by the retaining spring pressing against the post interior sides. Caps shall be removable without the use of tools. Construction shall be in accordance with 3.6.4 and finish shall be the same as for the bed ends (see 3.7).

3.5.3 Bed shoes. Bed shoes for closure of the bottom of the bed posts shall be as shown on figure 7. Bed shoes, four to be provided for each bed, shall be retained within the post by the retaining spring pressing against the post interior sides. Shoes shall be removable without the use of tools. Construction shall be in accordance with 3.6.5.

3.5.4 Adapters. When specified (see 6.2), each pair of beds shall be provided with a set of four adapters for double-decking two beds. Adapters shall be as shown on figure 13. Construction shall be in accordance with 3.6.6 and finish shall be the same as for the bed ends (see 3.7).

3.6 General construction. To insure interchangeability, beds shall be similar and of same construction in accordance with design, material thickness, dimensions and tolerances shown in figures 1 through 13 and specified herein. Materials required shall be in accordance with details specified in 3.4 and accessories specified in 3.5. Components for both size beds shall be the same, except the 84-inch long bed (Size 2) shall require two additional strands of spring fabric, side rails shall be 5-inches longer than the 79-inch long bed (Size 1), and spring fabric link sizes shall be in accordance with figure 4.

3.6.1 Bed end construction. The head and foot bed ends shall be interchangeable and of the same construction as shown on figure 8. Each bed end shall consist of two square tube posts, a rectangular horizontal top rail located between the posts, a full width panel located between the a channel at the top and a "Z" formation at the bottom, and two steel caps as closures at the top of each post, and two bed shoes as closures at the bottom of each post. The posts shall be 1-1/2 inch square steel tube with .4 gage thickness and shall be 30-1/2 inches high exclusive of caps and shoes. The horizontal top rail, channel, and "Z" formation thickness shall be of 14 gage thick steel. Connection of the two posts of each bed end shall be by placing the components in jigs and joining by the most applicable method of welding (see 3.9.1). The horizontal top rail shall be welded around the entire perimeter of each joint. The channel and "Z" formation shall be welded along the entire joint on the exposed side. Female corner lock shall be placed on main post as shown on figure 8 to match the male corner lock on the spring unit. The female corner lock shall be properly jigged and welded across the top, bottom, down one inch from top and up one inch from bottom as shown on figure 6. Finish of welds shall be in accordance 3.9.1. The end panel shall fit snugly into channel at the top. A chamfer as shown on figure 8 is permissable to

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ease installation. The bottom of the end panel shall rest on the horizontal section of the bottom "Z" formation and be secured with not less than three flat head sheet metal screws at locations shown on figure 8. Panels shall fill the full width between the posts.

3.6.2 Bed spring unit. The bed spring unit shall consist of an all welded angle frame, sinuous (no-sag type) spring fabric, two round tubular steel stretcher braces, and four male corner locks as shown on figures 2 and 3. The frame shall be constructed of steel angle sides and ends in accordance with 3.4.1.2 herein, with the 2-inch flange in the vertical position and the 1-1/2-inch horizontal flanges punched in accordance with figure 9 for insertion of spring fabric "Z" hook ends in side angles and spring helicals in the end angles. The horizontal flanges of the end angles shall overlay the horizontal flanges of the side angles with width and length as shown on figures 2 and 3. Frame shall be properly jigged and welded (see 3.9.1) as shown on figures 10 and 11. Two stretcher braces of steel tubing with 7/8-inch outside diameter and a 0.065 inch wall thickness shall be bowed, and the ends flattened as shown on figure 12. The stretcher braces shall be located on side angles as shown on figure 2 for Size 1 bed and as shown on figure 3 for Size 2 bed. Each end of the stretcher brace shall be secured to horizontal flange of side angle by a continuous weld along the entire union of the side angle surface and the stretcher brace edge. The male corner lock as shown on figure 5 shall be of the double stud design. The male corner lock plate shall be provided with two shouldered steel studs riveted (see 3.9.2) thereto to engage the female corner lock as shown on figure 5. The male corner lock shall be located on side angle as shown on figures 2 or 3 and welded thereto as shown figure 10. The upper and lower extension on the male lock end shall butt against the slotted surface of the female lock providing a lock wedge fit to prevent the bed from racking. There shall be four point contact between male and female corner locks; at upper and lower extensions on male lock end and the two studs. The spring fabric shall consist of 18 equally spaced transverse strands of no-sag type spring fabric for Size 1 bed and 20 strands for Size 2 bed. Each strand shall be connected to the adjacent strand with 5 rows of links (figure 4) clinched over the spring fabric. The "Z" hook ends of each strand shall be inserted in holes in horizontal flange of each side angle in a manner to provide secure retension, free from scratching and tearing hazard. Each end strand shall be attached to the end angle with five helical springs (figure 4).

3.6.3 End panel construction. End panels shall be of the dimensions shown on figure 8. The particleboard core shall be surfaced on each side with a one-piece high pressure decorative laminated plastic sheet. Adhesive for bonding the plastic sheet to the core shall conform to MMM-A-188 or MMM-A-180. The assembly periods, clamping, or pressing time shall be in accordance with the adhesive manufacturer's instructions to assure a permanent and secure bond. The upper horizontal edge of the panel may be chamfered to ease insertion into the channel. Panels shall extend the full width between the posts. Exposed surfaces of the panels shall not be cracked, split, scratched, scuffed, or dented.

3.6.4 Steel cap construction. The steel cap, consisting of a square washer and retaining spring, shall be fabricated as shown on figure 7. Each of the two spot welds shall resist 10-pound minimum pull. The washer shall be

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cold rolled steel, number 1 temper. The retaining spring shall be No. 1 full hard Rockwell B90 minimum. Caps shall be capable of being removed from the posts without the use of tools and shall require a 5-pound pull plus or minus 1-pound for removal.

3.6.5 Bed shoe construction. The bed shoe dimensional requirements shall be in accordance with figure 7 and assembly of components shall be as shown on the cross-section view shown on figure 7. The shoe base shall be cold rolled steel, AISI Designation C-1005, hardened to number 5 temper, and nickel plated. The interior of shoe base shall be covered with 1-3/4 inch diameter by 1/16 inch thick rubber washer to asure a tight base after curling operation.

The inside shell shall be cold rolled steel, die formed, and zinc plated. The center post shall be steel with AISI Designation C-12L14 and have a 1/2-inch hexagon head. The square washer on which the post rests shall be cold rolled steel number 1 temper with corners radius same as the posts and zinc plated. The retaining spring shall be cold rolled steel, 0.035-inch thick with a Rockwell hardness of B 88-93. The base shall be hardened and nickel plated prior to being curled. Slight cracks due to forming, hardening, and nickel plating is acceptable on the curled section of the base. Final assembly shall by swedging the corners of the center post hexagon head using a round punch. The square washer shall have same outside dimensions as the posts. Bed shoe shall be capable of being removed from the post without the use of tools and shall require a 10-pound pull plus or minus 1-pound for removal.

3.6.6 Adapter construction. The adapter dimensional and thickness requirements shall be in accordance with figure 13. The inner sleeve shall be a tight fit within the outer tube and further secured with two drive screws. The inner sleeve shall have a gap as shown on figure 13 to permit the sleeve to be squeezed for proper fit. Outer edges of the inner sleeve end shall be beveled to ease insertion into the main posts. Inserts shall be adjusted as necessary to fit snugly, after painting, into the top of the lower bed posts and the bottom of the upper bed posts. Finish shall be as specified in 3.7 and color shall be same as bed ends or as specified when procured separately (see 6.2). Adapters shall be tested for assembly and disassembly as specified in 4.4.2.

3.7 Finish. All metal surfaces of the bunk beds, except plated steel shoes and panels, shall be painted. Application of finish shall produce smooth and uniform surfaces without runs, wrinkles, or areas of thin or no film and free from objectionable defects.

3.7.1 Surface preparation. All surfaces to be painted shall be treated in accordance with type I or type II of TT-C-490 except phosphate coating may be applied in three stages and a non-chromated final acidic rinse maybe used.

3.7.2 Painting. All treated surfaces shall be painted with an epoxy-power coating of color specified (see 3.4.5 and 6.2) applied to a minimum dry film thickness of 1.1 mil. The coating baking period shall be as recommended by the coating manufacturer.

3.8 Marking. The letters "US" in characters 1-inch high shall be indented at center of horizontal or vertical surface on one end angle of each bed. The specification number, stock number, nomenclature, manufacturer's

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[illegible] trade name or trademark of such character easily identifiable with the manufacturer, and the year and contract number shall be indented or on an aluminum plate permanently attached on same end angle surface and to right of indented characters "US". Bed ends manufacturer shall be indentified by identing the manufacturer's initials or trademark of such character easily identifiable with the manufacturer on one surface of one or both female corner locks. Letters or trade mark shall be approximately 1/2-inch high at location shown on figure 6.

3.9 Workmanship. Workmanship shall be of the highest quality throughout and in accordance with the best standard practice used in the manufacturer of high grade commercial furniture of this type.

3.9.1 Welding. All welds shall be sound, and smooth, free from pits, holes, and fissures. Fillet welds shall be in accordance with the definition and size requirements of AWI A3.0. The deposited weld metal shall thoroughly fused with base metal and edges of union and shall penetrate into the root of the joint. All rough surfaces or projecting fins shall be ground smooth to prevent personal injury or damage to bedding. Weld spatter or flux shall be removed from surfaces surrounding the welding area.

3.9.2 Riveting. Studs shall be secured to the male corner locks by peening as shown on figure 5. The studs shall be tight with the shoulder and end in full contact with the male corner lock surfaces. The head of the stud end shall be flat, parallel to the male corner lock surface, and edges free of burrs or fins.

3.10 Assembly instruction. Prior to shipment, each shipping container (carton) shall contain assembly instruction for single bed, assembly and double decking (see 6.4). In addition, a sketch similar to figure 14 shall be included in instructions to preclude the possibility of forcing male corner lock studs to seat or bottom in female corner lock. Instruction shall be provided on a white paper sheet 8-1/2 inches wide by 11-inches long.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless othewise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. This inspection requirements specified herein are classified as follows.

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, when specified (see 6.2), and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling shall be in accordance with MIL-STD-105. A lot shall consist of one day's production of bunk beds or adapters, but shall not consist of more than 500 bunk beds. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level S-4 and an Acceptable Quality Level (AQL) of 2.5 defects per hundred units (DHU) for major defects and 4.0 DHU for minor defects.

4.3.2 Sampling for tests. Test shall be based on inspection level S-4 and an AQL of 2.5 DHU.

4.4 Examination. Each sample shall be examined for compliance with the requirements in Section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.4.1 Examination. Each sample shall be examined for defects listed in table I, 4.4.1, and 4.4.2. Any sample having one or more of the defects listed, as applicable, shall considered a defective unit.

TABLE I. Classification of defects.

Categories	Defects	Requirement
Critical	None defined.	
Major	(In process examination)	
101	Steel surfaces not as specified.	3.4.1
102	Surfaces to be painted not prepared as specified.	3.7.1
104	Coating baking cycle not as specified.	3.7.2.

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Major	(End item examination)	
105	Design and construction not as specified.	3.1
106	Overall dimensions not as specified.	fig. 1
107	Parts not manufactured to interchangeable standards.	3.2 and 3.6
108	Steel components not as specified.	3.4.1.1 thru 3.4.1.6
109	End panel core not as specified.	3.4.2
110	High pressure decorative laminated plastic sheet not as specified.	3.4.3
111	Adhesive not as specified.	3.4.4
112	Coating not as specified.	3.4.5
113	Plastic sheet not on both sides of panel.	3.5.1
114	End panels not as specified.	3.5.1
115	Steel caps not as specified.	3.5.2
116	Bed shoes not as specified.	3.5.3
117	Adapters not as specified.	3.5.4
118	Bed end dimensions not as specified.	3.6.1
119	Steel components not thickness specified.	3.6.1
120	Welding of bed and horizontal components not as specified.	3.6.1
121	Welding of female corner lock not as specified.	3.6.1 & fig. 6
122	End panels not installed as specified.	3.6.1
123	Bed spring unit angle frame members not welded as specified.	3.6.2 and figs. 10 and 11
124	Long leg of angle sides not in vertical position as specified.	3.6.2
125	Stretcher braces not as specified.	3.6.2
126	Stretcher braces not located as specified.	3.6.2
127	Corner locks not located as specified.	3.6.2
128	Number of strands of spring fabric not as specified.	3.6.2
129	Spring fabric loose and not attached as specified.	3.6.2
130	Hook ends of spring fabric not free of scratching and tearing hazard.	3.6.2
131	End panel construction not as specified.	3.6.3
132	Steel cap construction not as specified.	3.6.4
133	Bed shoe construction not as specified.	3.6.5
134	Bed shoe steel components not of material specified.	3.6.5
135	Outside dimensions of bed shoe washer and post not same.	3.6.5
136	Adapter contruction not as specified.	3.6.6
137	Finish not without runs, wrinkles, or areas of thin or no film, or other defects such as indication of rust or scale under finish or discoloring.	3.7
138	Coating not of thicknesses specified.	3.7.2
139	Welding not as specified.	3.9.1
140	Studs not secured as specified.	3.9.2
141	Assembly instructions not provided.	3.9.10

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Minor:

201	Edges of angles not rounded.	3.4.1.2
202	Color not as specified.	3.4.3
203	When woodgrain end panels are specified, woodgrain not in horizontal direction.	3.5.1
204	Pull necessary to remove steel caps not as specified.	3.6.4
205	Pull necessary to remove steel shoes not as specified.	3.6.5
206	Identification marking missing.	3.8
207	Assembly instructions not provided.	3.10

4.4.1 Bed assembly examination. Sample beds from each inspection lot shall be assembled to determine ease of assembly and proper fit of mating parts. In addition, the beds shall be double-decked to assure proper overall dimensions among beds. Failure of studs and male corner lock extensions (see 3.6.2) to fit properly, requiring excessive force or use of tools to assemble or inability to match top bed post bottoms and bottom bed top when double-decking each shall consititute a major defect.

4.4.2 Adapter assembly examination. When adapters are procured separately or with beds, samples shall be examined for ease of double-decking. Assembly and disassembly with beds or steel tubes of same dimensions and wall thickness as bed posts shall be performed manually without the use of tools and shall be a snug fit. Inside dimensions of bed posts or steel tubes shall be checked periodically for proper dimensions (see figure 13).

4.5 Tests.

4.5.1 Metals. Specimens of metal components shall be tested in accordance with FED-STD-151, Supp. A and B, wherever applicable.

4.5.2 Bed test. A 600 pound load shall be placed across the side angles of spring unit of an assembled bed, midway between the bed ends. The test load bottom or base shall be approximately 18-inches wide. At the center of the spring unit, the deflection of the spring unit side angles shall be not more than 7/16 inch with the 600 pound load in place. Upon removal of the test load, the spring unit shall return to its original position with no visible distortion in the bed locks or spring unit frame.

4.6 Packing inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of Section 5. The inspection shall be in accordance with MIL-STD-105. A completed pack prepared for shipment shall be considered a unit of product. A lot shall consist of all completed packs prepared for shipment during any one day period. The inspection level shall be level II with an AQL of 4.0 percent defective.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

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5.1.1 Level A. The main posts, horizontal tube, and end panel of the bed ends (head and foot ends), and the end angles and corner locks of the bed spring unit shall be completely wrapped with cushioning material securely fastened in place with tape or twine in a manner which will prevent shifting or becoming detached. Additional cushioning material shall be used as required to protect any other metal surfaces from damage. One complete single bunk bed shall be arranged into a compact assembly in order to provide the smallest practicable shipping cube and to provide support for the faces and edges of the exterior shipping container. The assembly shall be secured in position by the application of flat steel straps at each end which completely encircle the main posts and the end angles. Each strap shall be drawn as tight as possible to provide a compact non-shifting assembly. The strapping shall be minimum 3/4-inch width and shall be prevented from coming in direct contact with metal surfaces of the assembly by means of the applied cushioning or by use of additional separating material. When adpaters are furnished with the bunk beds (see 3.5.4), they shall be cushioned with the materials specified herein and secured in an appropriate location inside the bed spring units with filament reinforced tape or flat steel strapping.

5.1.1.1 Adapters (procured separately). Adapters procured separately from bunk beds shall be packaged in quantities of four in a close-fitting fiberboard box of minimum 175 psi bursting strength and the box closed and secured. The adapters shall be cushioned to prevent contact between pieces.

5.1.2 Level C. Bunk beds (with adapters when furnished) and adapters (when furnished separately) shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. A complete single bunk bed, packaged as specified in 5.1.1, shall be packed in a close-fitting box conforming to PPP-B-601 overseas type or PPP-B-621, class 2. The box shall be close-fitting such that the cushioning material on the outermost parts of the assembly bears against the inner surfaces of the exterior shipping container in a snug manner. The box shall be provided with a waterproof case liner conforming to MIL-L-10547, sealed in accordance with the appendix thereto.

5.2.1.1 Adapters (procured separately). Adapters procured separately from bunk beds, packaged as specified in 5.1.1.1, shall be packed in boxes specified in 5.2.1. The boxes shall be provided with case liners as specified in 5.2.1.

5.2.2 Level B. A complete single bunk bed, packaged as specified in 5.1.1, shall be packed in a close-fitting box conforming to either PPP-B-640, class 2, PPP-B-601, domestic type, or PPP-B-621, class 1. PPP-B-640 boxes shall be closured with flat steel strapping and waterproofed with tape in accordance with the appendix to the box specification.

5.2.2.1 Adapters (procured separately). Adapters procured separately and packaged as specified in 5.1.1.1 shall be packed as specified in 5.2.2 or in boxes conforming to PPP-B-636, class weather-resistant and closed in accordance with Method V of the appendix thereto.

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5.2.3 Commercial. Packaged bunk beds (with adapters when furnished) and packaged adapters (when furnished separately) shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Boxes containing both bunk beds and adapters shall be marked noting that adapters are also included in the box.

6. NOTES

6.1 Intended use. This specification covers square tube bunk beds with end panels for use in unaccompanied enlisted personnel housing.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Whether first article is required (see 3.3).
d. Specify the pattern, color, and finish of high pressure laminated plastic sheets required for panels (see 3.4.3).
e. Specify color required for bed end and spring unit (see 3.4.6).
f. Whether each pair of single beds shall be supplied with a set of four adapters (see 3.5.4 and 3.6.6).
g. Whether testing is required during quality conformance inspection (See 4.2.2).
h. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Nomenclature. The approved nomenclature of the item covered by this specification is as follows:

Bed, Bunk, Steel, Square Tube with End Panels, Single and Double-Deck.

6.4 Bunk bed assembly instructions.

Single bed assembly instructions. Assemble a single bunk bed by placing a spring unit on the floor with spring up, stand a bed end at each end of the spring unit, with one person standing outside of each bed end, lift the spring unit up and place between the slotted extensions on the main posts of the bed end and fit the studs into to the slots. Lightly tap each corner to set the studs in the slots. DO NOT FORCE THE STUDS TO SEAT AT BOTTOM OF THE SLOTS. See illustration below.

Double decking instructions. Assemble two bunk beds as instructed above. Remove the caps from the top of the main posts at each corner of one bed and remove the bed shoes from the bottom of the main posts of the other bed. Place an adapter into each post from which the caps were removed. Lift the other bed above the adapters and insert the top of the adapters into the bottom of the main posts.

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6.5 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of one bed or two if adapters are specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

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Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

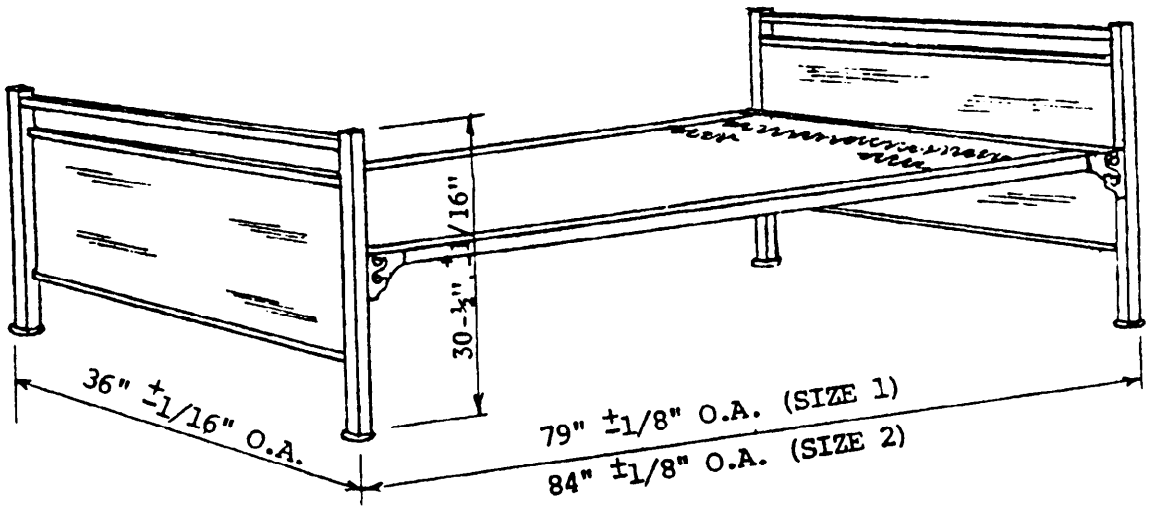
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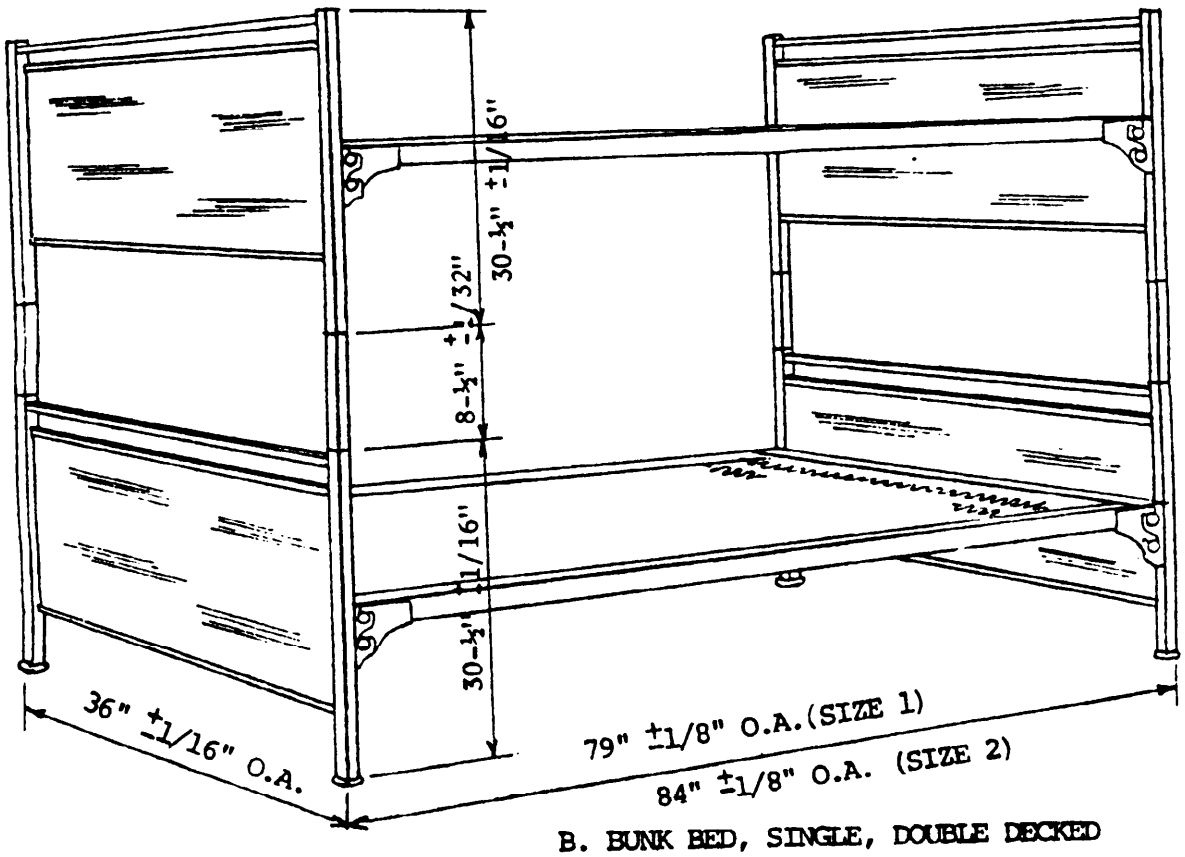
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NOTE: MEASURE OVERALL BUNK BED LENGTH AT MID POINT BETWEEN TOP AND BOTTOM OF BED POSTS.



A. BUNK BED, SINGLE

DO NOT SCALE



B. BUNK BED, SINGLE, DOUBLE DECKED

FIGURE 1. BUNK BED, STEEL, SQUARE TUBE WITH PANELS, SINGLE AND DOUBLE DECK

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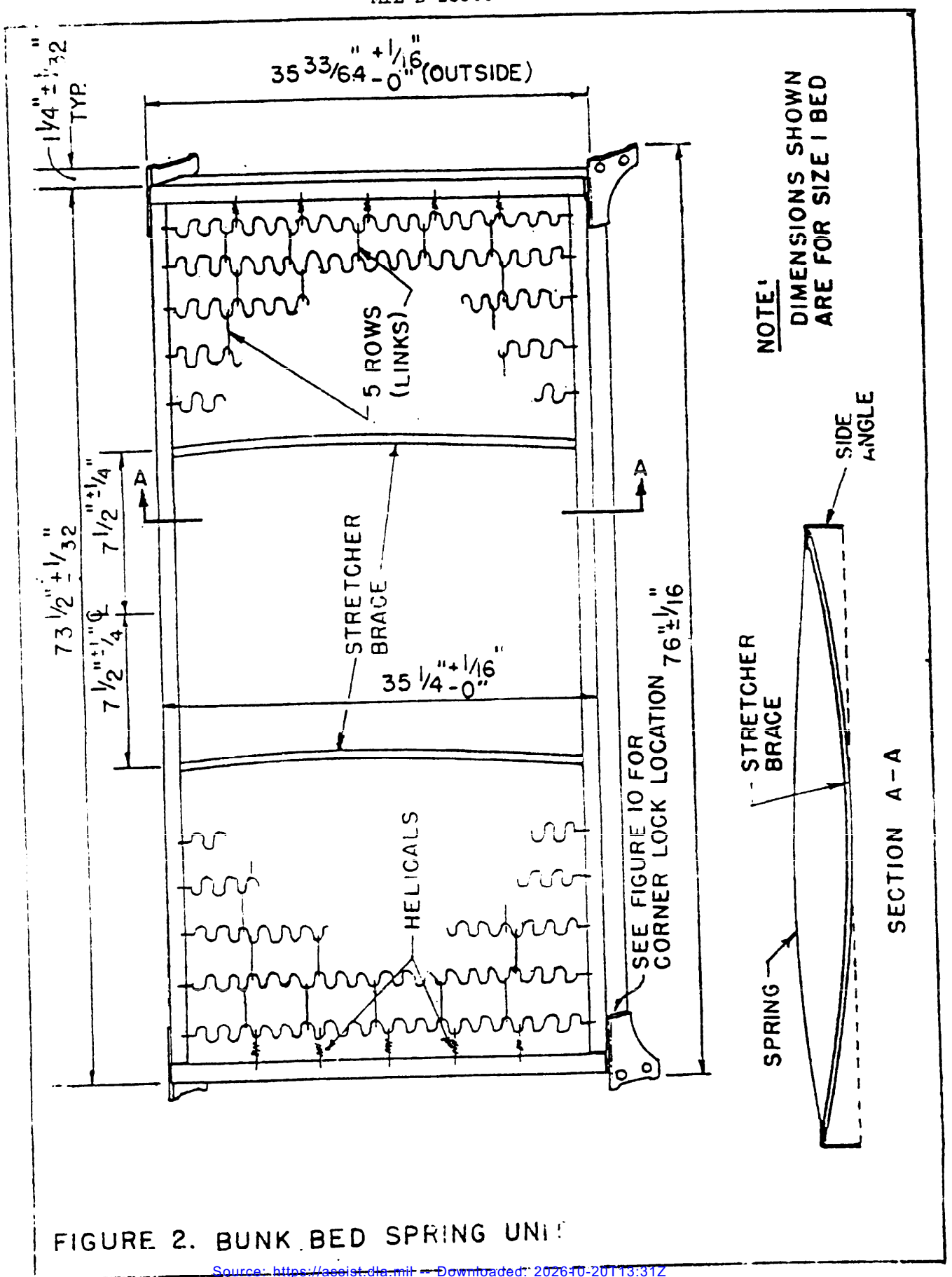


FIGURE 2. BUNK BED SPRING UNIT

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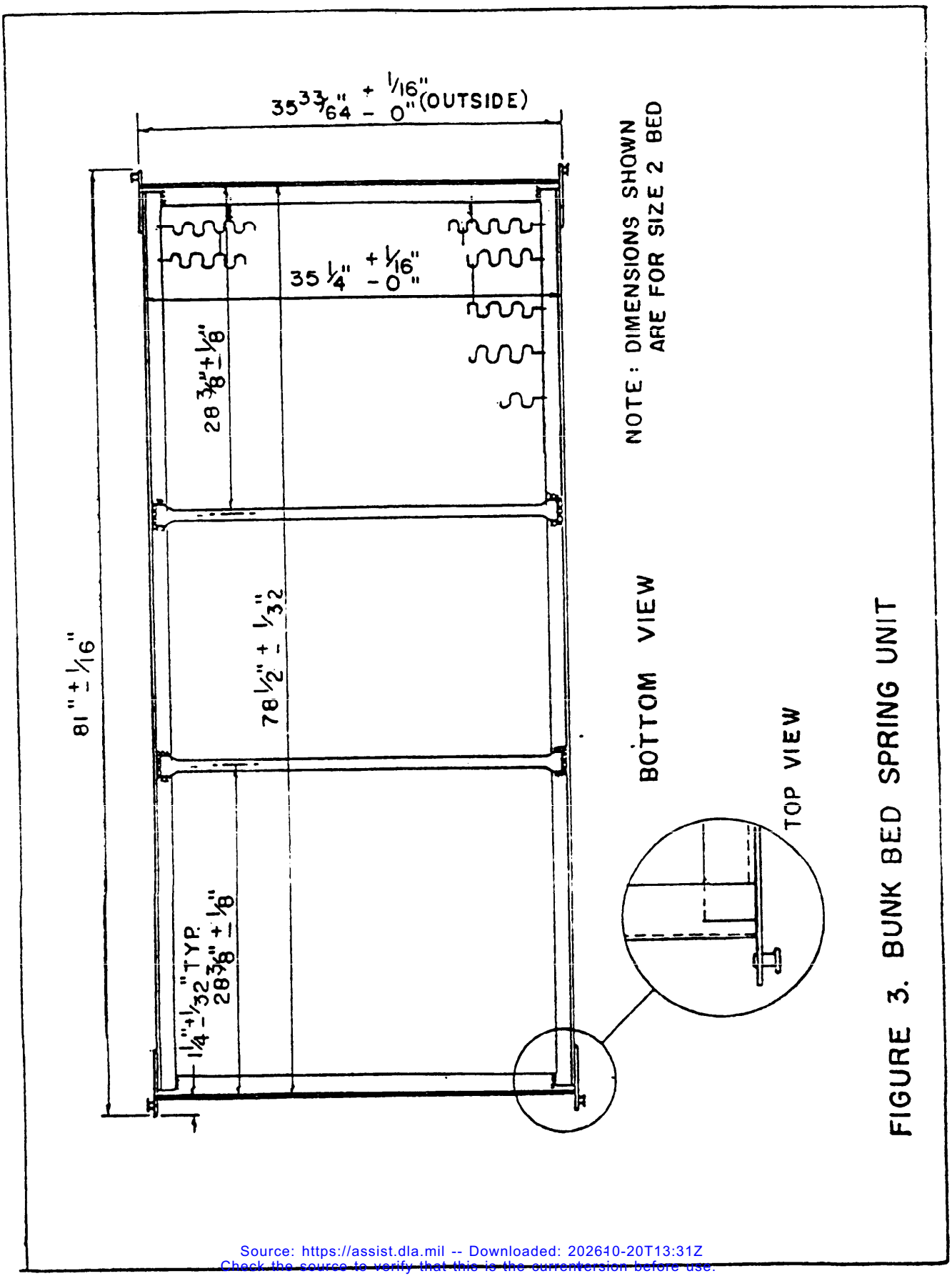
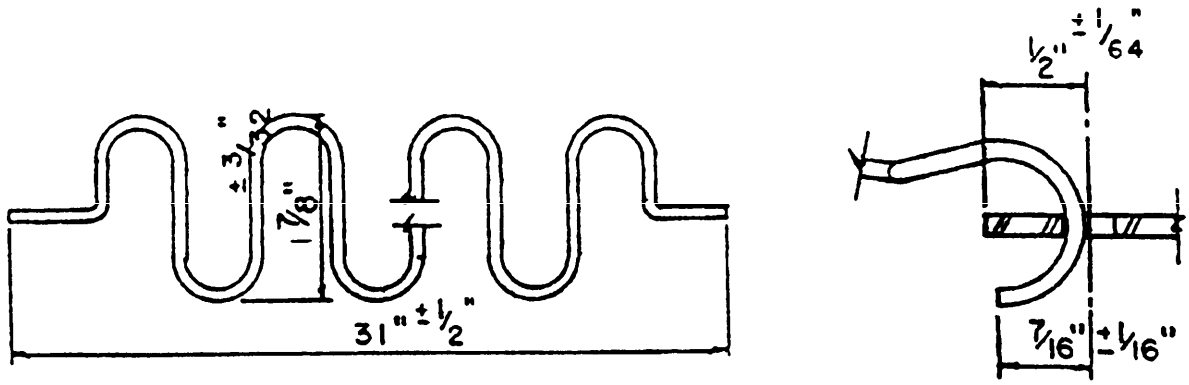
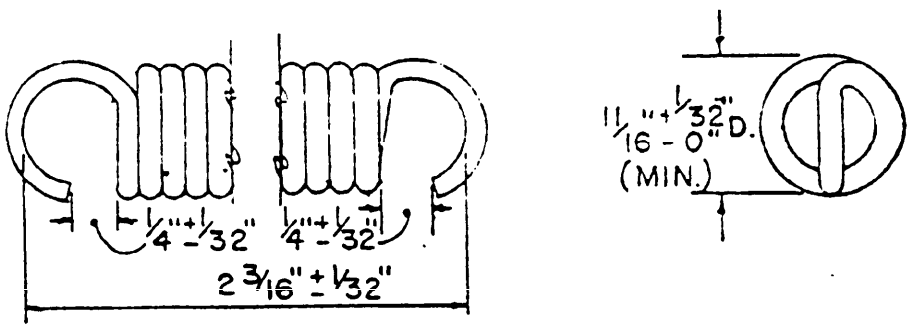


FIGURE 3. BUNK BED SPRING UNIT

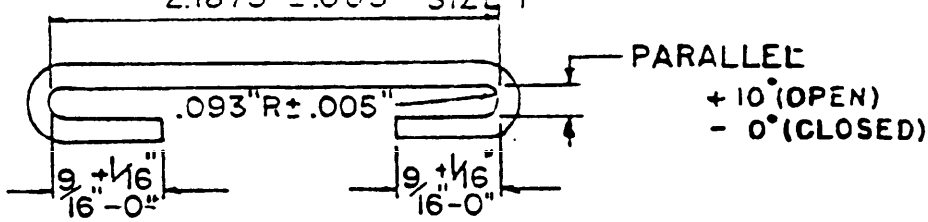
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A. SPRING FABRIC (no sag type)



B. SPRING FABRIC HELICALS
(12 turns close wound)
2.045" ±.005" SIZE 2
2.1875" ±.005" SIZE 1



C. SPRING FABRIC LINK

FIGURE 4. SPRING UNIT COMPONENTS

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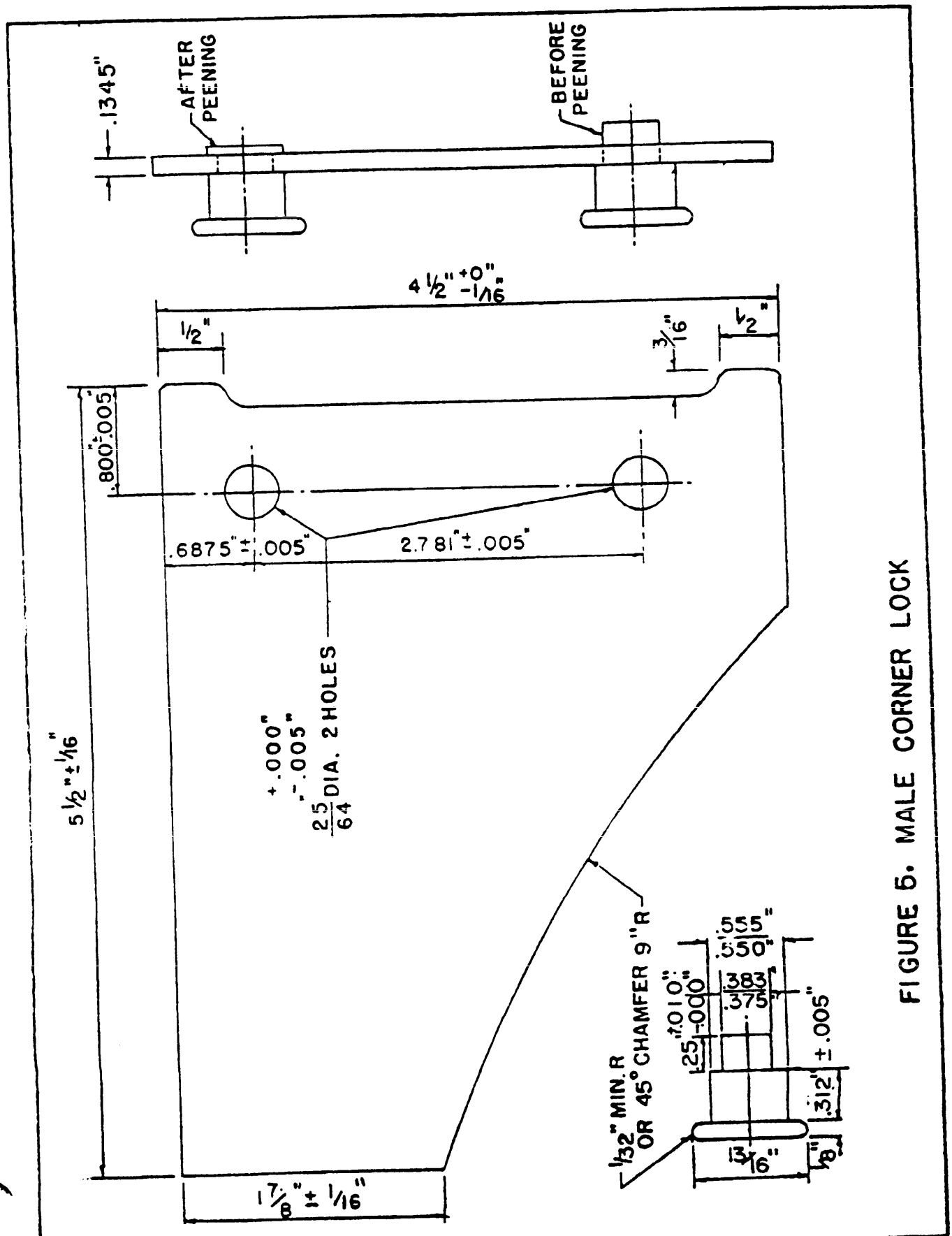


FIGURE 6. MALE CORNER LOCK

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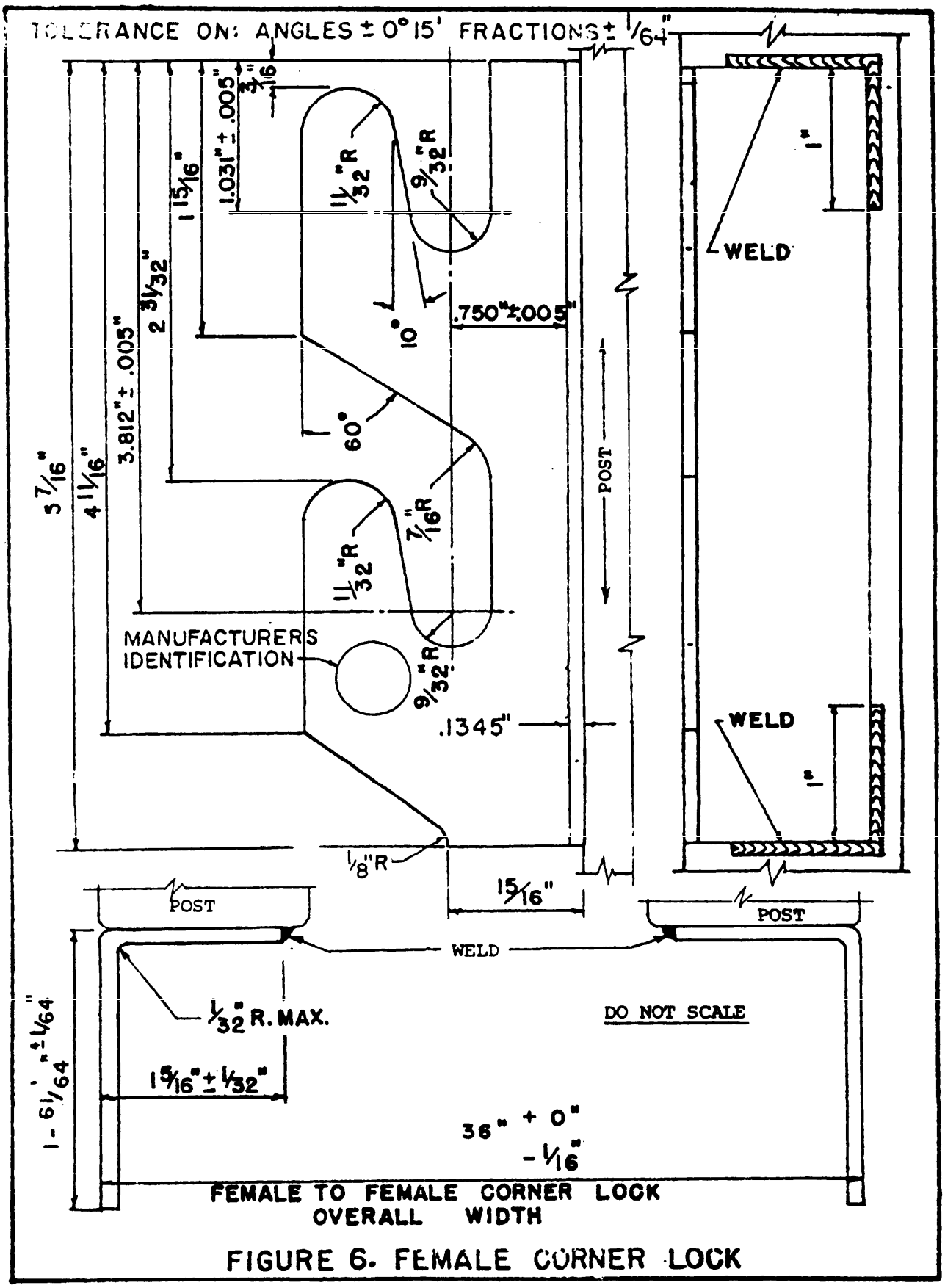


FIGURE 6. FEMALE CORNER LOCK

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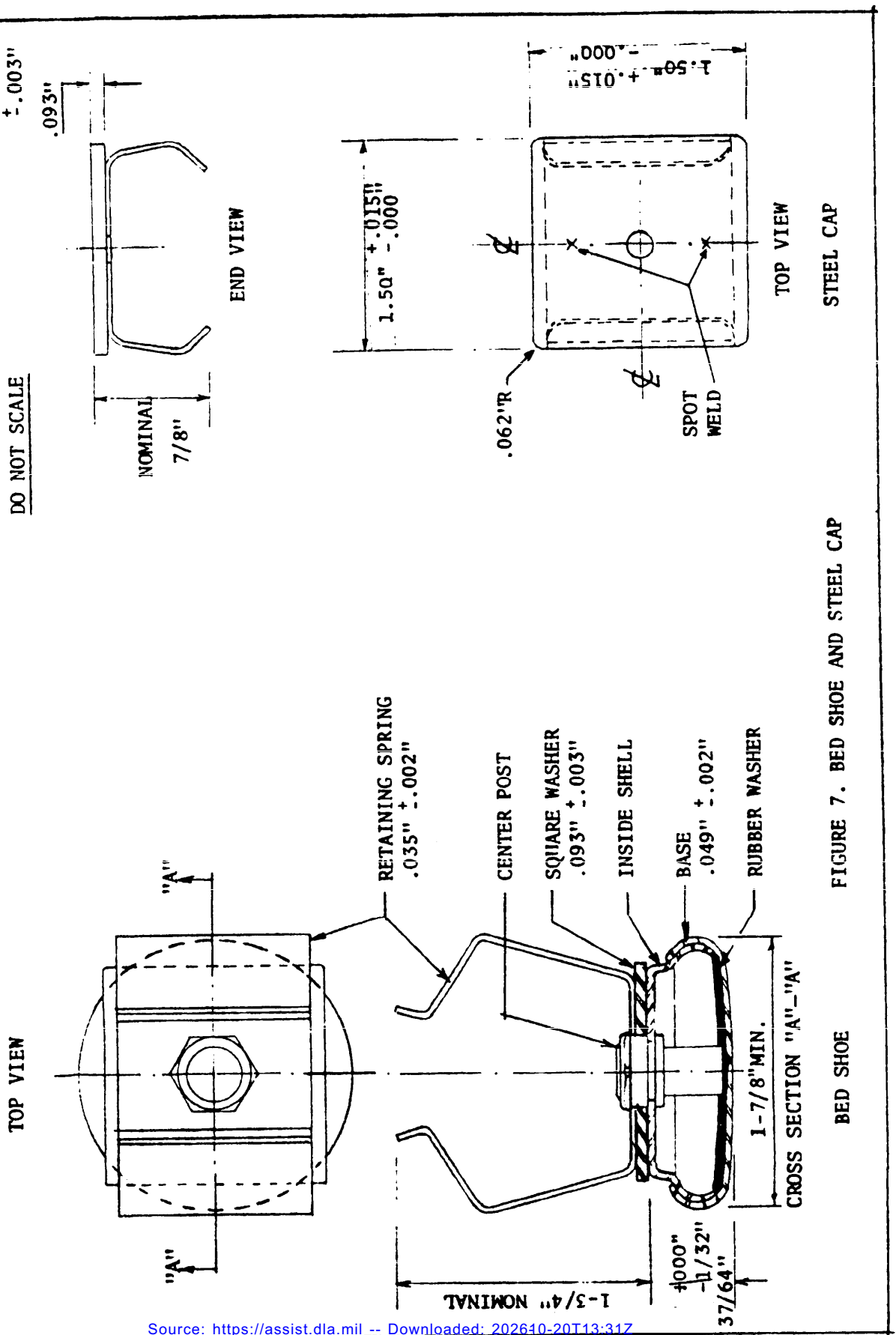


FIGURE 7. BED SHOE AND STEEL CAP

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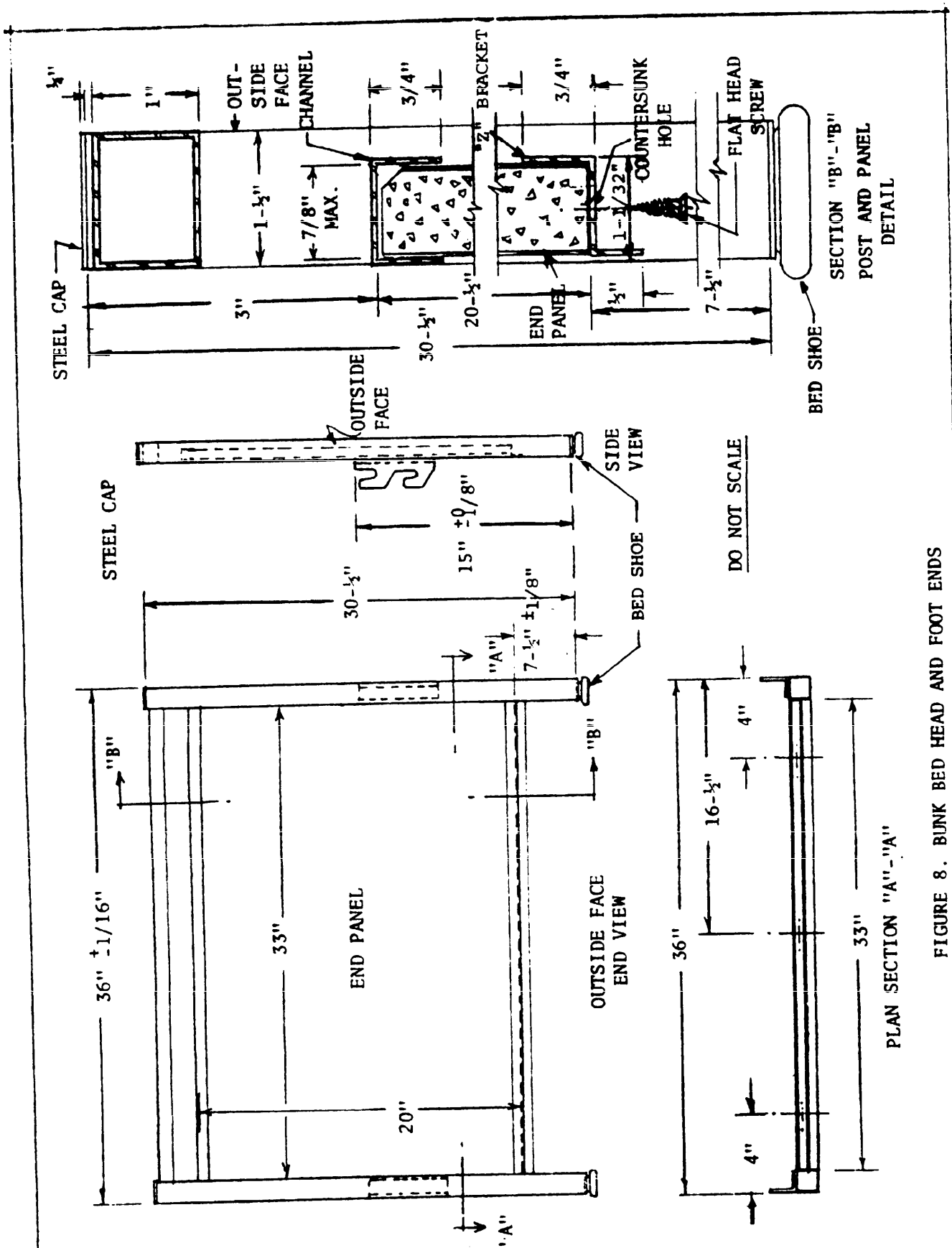


FIGURE 8. BUNK BED HEAD AND FOOT ENDS

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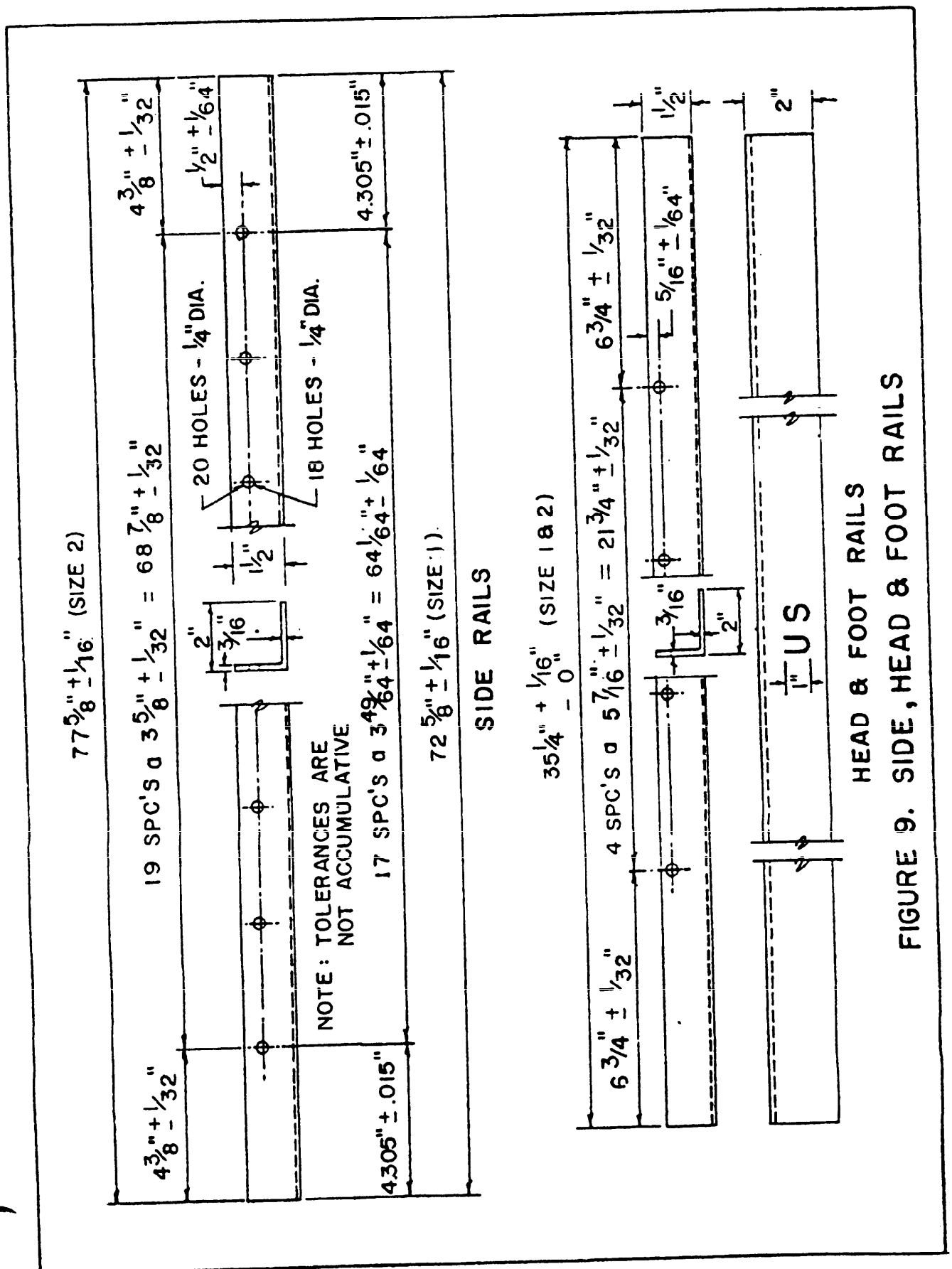


FIGURE 9. SIDE, HEAD & FOOT RAILS

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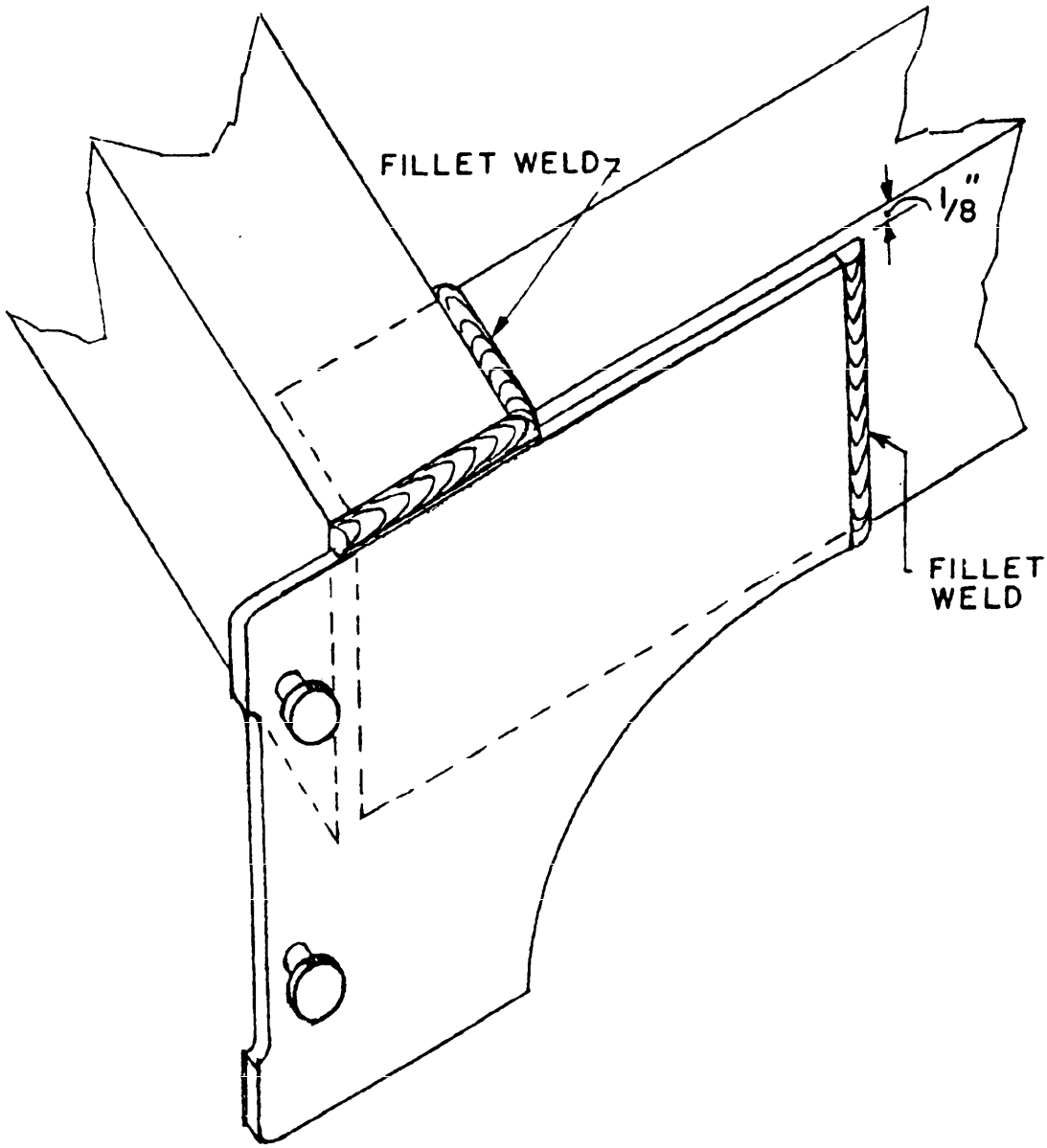


FIGURE 10. SPRING FRAME WELDING, TOP VIEW .....

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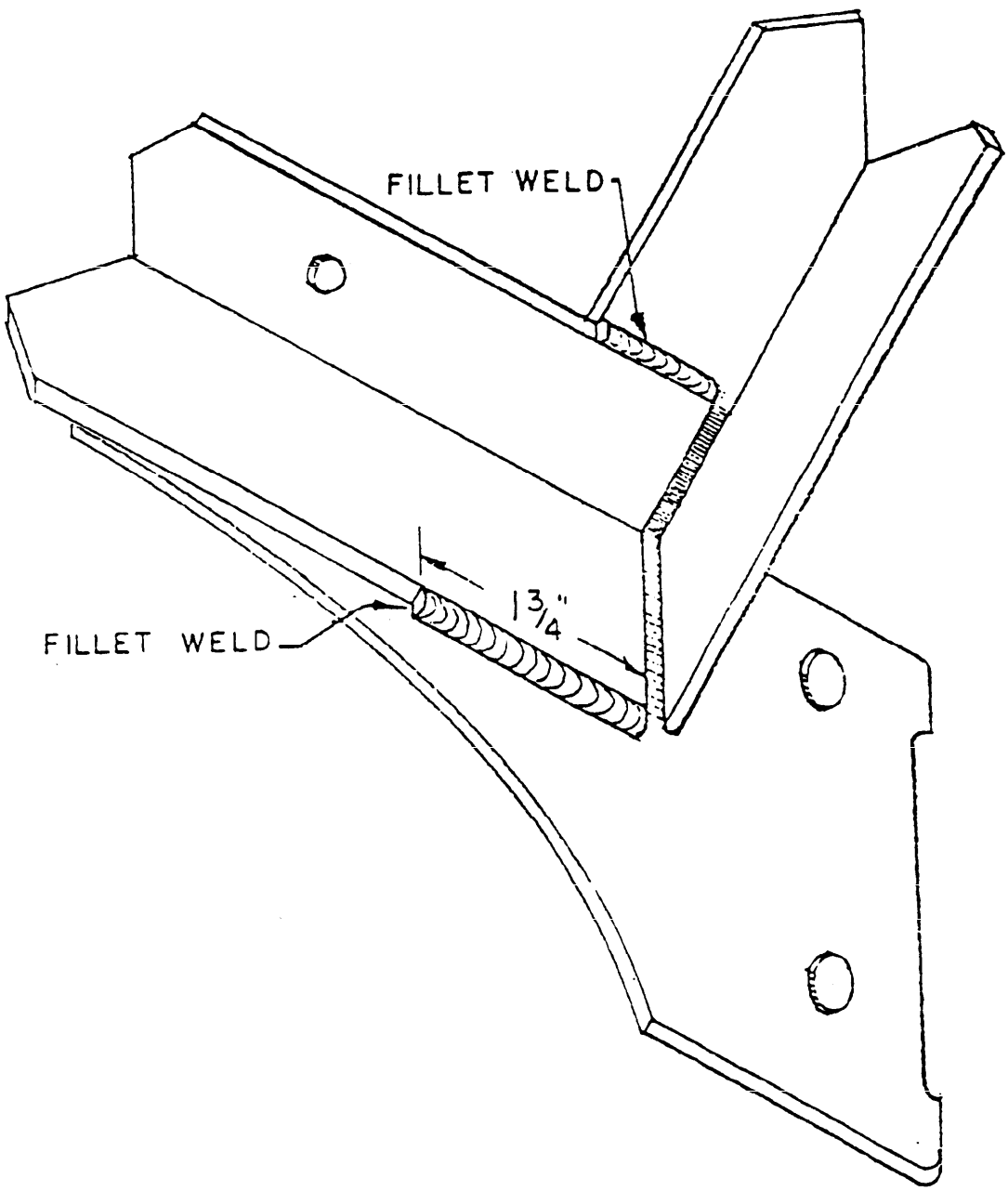


FIGURE 11. SPRING FRAME WELDING, BOTTOM VIEW .....

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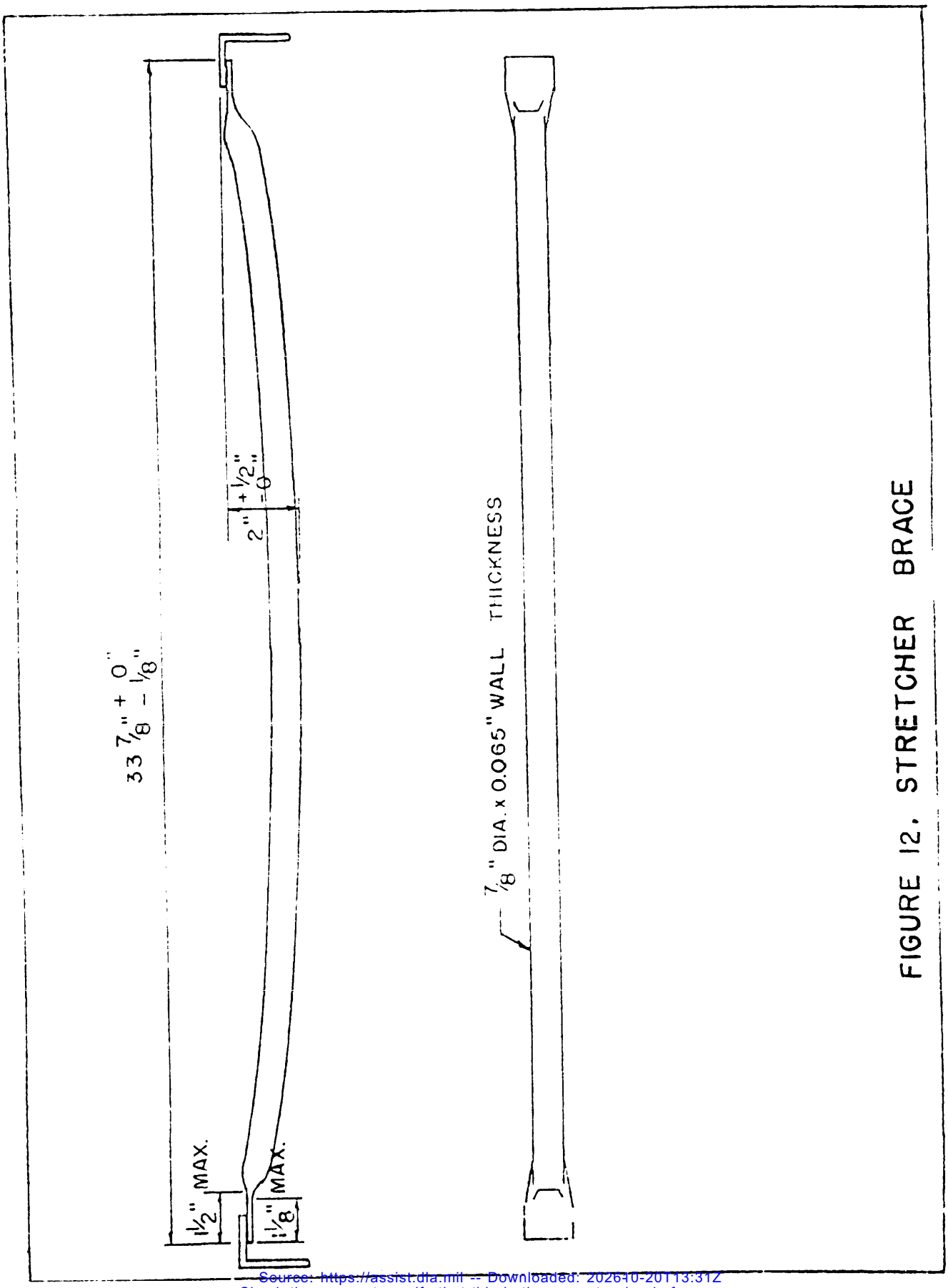


FIGURE 12. STRETCHER BRACE

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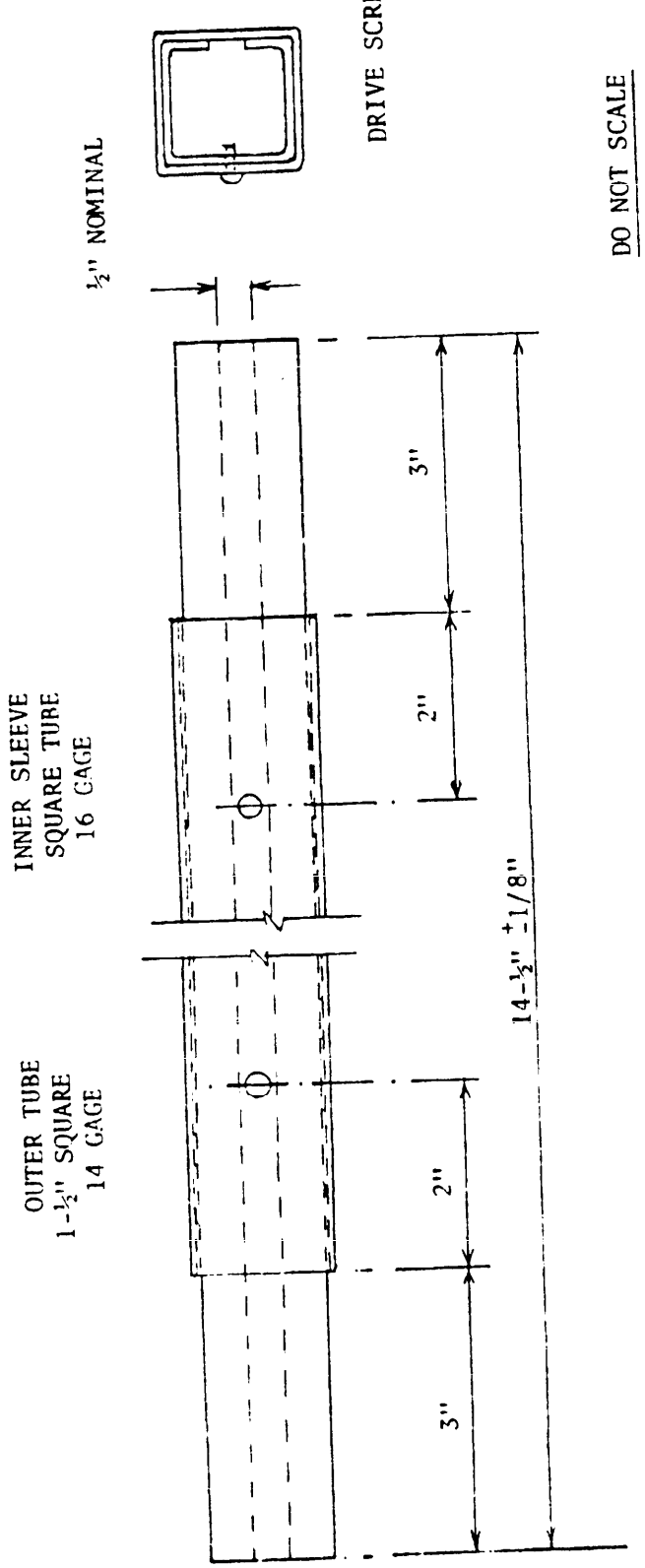


FIGURE 13. ADAPTER, SQUARE TUBE, FOR CONVERSION OF SINGLE BEDS TO DOUBLE DECK

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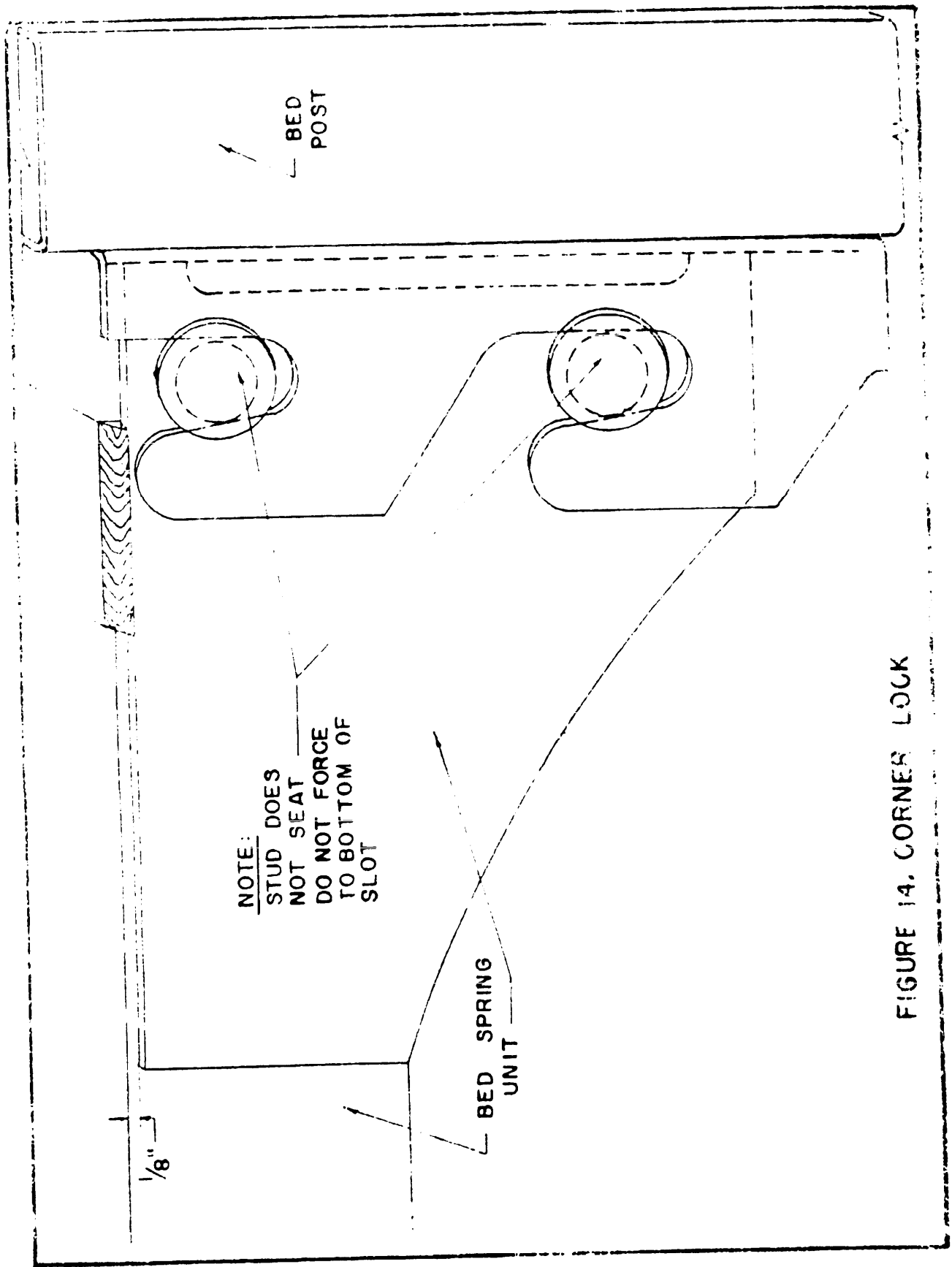


FIGURE 14. CORNER LOCK

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